

Companion LXXXIII: Fisher Forecasts for α Using Transport-Based Topological Statistics in the Relaxation-Driven Cyclic Cosmology

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Abstract

Transport-based statistics provide a powerful way to quantify differences between persistence-diagram ensembles across cosmological models. Building on the unified diffusion-optimal transport framework developed in previous Companions, this work constructs Fisher forecasts for the relaxation parameter α in the Relaxation-Driven Cyclic Cosmology (RDCC). The key observable is a scale-resolved transport discrepancy between RDCC and Λ CDM persistence diagrams, computed using Wasserstein, Sinkhorn, or neural transport maps. We derive the Fisher information associated with these transport statistics and construct α -sensitivity kernels that identify the angular scales and topological features most informative for constraining α in Euclid-like weak-lensing surveys. This Companion provides the first explicit link between geometric transport quantities and forecasted RDCC parameter constraints.

1 Introduction

Persistence diagrams encode the topological structure of weak-lensing convergence fields. Previous Companions introduced a hierarchy of geometric tools for these diagrams, including Wasserstein distances, entropic optimal transport, Schrödinger bridges, score-based diffusion models, and neural transport maps. Companion LXXXII introduced transport-based parameter inference and α -sensitivity maps.

The present Companion takes the next step: we construct Fisher forecasts for the RDCC relaxation parameter α using transport-based topological statistics. This provides a direct and quantitative link between geometric transport quantities and observable constraints from Euclid-like surveys.

2 Transport-Based Observables

Let $\mathcal{D}_\Lambda$ and $\mathcal{D}_\alpha$ denote persistence-diagram ensembles from Λ CDM and RDCC with parameter α . We define a scale-resolved transport statistic

$$\mathcal{T}_\ell(\alpha) = \mathbb{E}_{D \sim \mathcal{D}_\Lambda} [W_p(D, T_\alpha(D))]_\ell, \quad (1)$$

where the subscript ℓ indicates restriction to features associated with angular multipole ℓ . The neural transport map T_α is trained to push $\mathcal{D}_\Lambda$ towards $\mathcal{D}_\alpha$.

3 Fisher Information for α

Given a vector of transport statistics

$$\mathbf{T}(\alpha) = \{\mathcal{T}_\ell(\alpha)\}_\ell,$$

with covariance matrix $\mathbf{C}$ estimated from mocks, the Fisher information for α is

$$F_{\alpha\alpha} = \left(\frac{\partial \mathbf{T}}{\partial \alpha}\right)^T \mathbf{C}^{-1} \left(\frac{\partial \mathbf{T}}{\partial \alpha}\right). \quad (2)$$

The forecasted constraint is

$$\sigma(\alpha) = F_{\alpha\alpha}^{-1/2}.$$

The derivative $\partial \mathbf{T} / \partial \alpha$ is estimated using RDCC mocks at nearby α values.

4 α -Sensitivity Kernels

We define the ℓ -resolved sensitivity kernel

$$K_\alpha(\ell) = \left[\frac{\partial \mathcal{T}_\ell}{\partial \alpha}\right]^2 / C_{\ell\ell}. \quad (3)$$

This kernel quantifies the contribution of each angular scale ℓ to the total Fisher information.

Large values of $K_\alpha(\ell)$ indicate scales where RDCC-induced topological deformations are most pronounced.

5 Connection to Euclid-Like Surveys

For a Euclid-like survey, the covariance matrix $\mathbf{C}$ is dominated by:

- shape noise,
- finite sampling of galaxies,
- cosmic variance,
- reconstruction noise in convergence maps.

Transport statistics are robust to these effects because:

- Wasserstein/Sinkhorn distances are stable under noise,
- neural transport maps can be trained to denoise diagrams implicitly,
- diffusion-based models provide uncertainty-aware transport.

Thus the Fisher forecasts derived here are directly applicable to Euclid mocks.

6 Conclusion

Transport-based topological statistics provide a powerful and robust way to quantify differences between RDCC and Λ CDM. By constructing Fisher forecasts for the relaxation parameter α , we establish a direct link between geometric transport quantities and observable constraints. The α -sensitivity kernels introduced here identify the angular scales most informative for RDCC parameter inference in Euclid-like surveys. This Companion thus completes the bridge from geometric theory to observational forecasting.

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